

Neo v2 — Pilot Deployment to VPS

Status report · Jenner Ramos · 18 August 2026

Status: live. The Neo pilot has been moved off the local workstation and is running on a dedicated cloud server since 17 August 2026, with the full board-meeting corpus migrated, password protection in place, nightly backups running, and a question answered in about 5–8 seconds end to end.

1. HOW IT WAS DONE

Six steps, in order. The first two were prerequisites: Neo originally ran its own local language model, which no small server could host.

#	Step	What it changed
1	Made the answering model swappable 13 Aug	A provider layer was added so the question-answering endpoint can use any hosted model (Gemini, OpenAI, Anthropic, DeepSeek, or the local one) by changing a setting rather than code. Generation became an outbound web call — <i>this is what removed the need for a GPU server</i> and made a small CPU machine viable. Running on Gemini Flash; answer quality on the graded set went up, not down.
2	Made the ingestion model swappable 14 Aug	The transcript-extraction steps (votes, financials, personnel, initiatives) were moved onto the same layer, with their <i>own separate</i> settings so ingestion can never silently follow the answering model onto a paid API. Model choice was decided by scoring candidates against transcript-verified items, not by preference. This is what will later allow ingestion to run off the workstation too.
3	Built the container images 16 Aug	Two images, both multi-stage for size: a backend image (CPU-only build — the GPU version would have added ~3 GB of unused drivers — with the search models baked in so a restart needs no downloads) and a frontend image built in standalone mode. Sized so one worker fits comfortably in memory.
4	Assembled the stack 16 Aug	A Compose file wires the five services together with a reverse proxy in front, the database and vector index kept private, password protection at the edge, and a single settings file per environment. Rehearsed locally end to end before touching a server.
5	Provisioned and cut over 17 Aug	Server rented and hardened, stack deployed, and the real data copied across (section 3).
6	Measured and tuned 17–18 Aug	First live timings were too slow to pilot; fixed by changing the re-ranking model (section 4). Nightly backups added and a restore drill run.

2. WHAT RUNS ON THE SERVER

Five services behind a single front door. Nothing except the reverse proxy is reachable from the internet.

Service	Role
Caddy (proxy)	The only public port. Routes traffic, enforces the pilot password, and keeps live-streamed answers un-buffered.
Web (Next.js)	The user interface.
API (FastAPI)	Search, retrieval and the <code>/ask</code> endpoint. Text generation is an outbound call to the hosted LLM, so no generation model sits on the server.
Postgres	Meetings, transcripts, votes, financials, extracted initiatives.
Qdrant	Vector index of the transcript corpus.

- Access control is a single pilot username and password applied at the proxy; the health check stays public so the server can be monitored without credentials.
- One environment file holds every secret and setting for the server, kept separate from the workstation configuration so the two can never be mixed up.

3. SERVER AND CUTOVER

Host	OVH cloud VPS, Ubuntu 24.04, 4 vCPU / 7.7 GB RAM / 72 GB disk
Address	<code>http://15.204.248.52</code> (IP only; no domain name yet)
Data migrated	Full database dump (39 MB) plus the vector index snapshot (897 MB, 98 MB compressed)
Verified after restore	12,464 indexed chunks, index status green, and every table count matching the workstation exactly

The migration deliberately moved *real data* rather than rebuilding from scratch, so the pilot server is an exact copy of what had already been reviewed locally.

4. PERFORMANCE WORK (THE MAIN ENGINEERING WIN)

The first live measurements showed a question taking **48–57 seconds** — almost all of it spent re-ranking search results on the CPU. That is not usable for a pilot. Replacing the re-ranking model with an ONNX version cut it dramatically, with no loss of answer quality.

Measured on the live server	Before	After
Retrieval + re-ranking	~40 s	2.4–5.8 s
Total time per question	48–57 s	4.6–7.9 s
Memory used by the API	2.5 GB	956 MB
Deployed image size	6.96 GB	3.48 GB
Evaluation suite (33 graded questions)	32 / 33	33 / 33

Quality was checked before switching: both versions were run side by side on the same server against the same 33-question evaluation set and surfaced the same supporting evidence on every case that could be measured. The previous configuration is kept on the server for a one-line rollback.

5. BACKUPS AND SAFETY

- Nightly automated backup (03:30 UTC) of the database and the proxy certificate store; 14 daily and 4 weekly copies retained.
- Backups are validated on write — a truncated dump is rejected rather than silently kept.
- A full **restore drill was performed and passed**: the newest backup was restored into a scratch database and every table matched the live system.
- The vector index is intentionally not backed up nightly — it is rebuildable from source data, and would otherwise dominate the disk.

6. VERIFIED ON THE LIVE SERVER

- Health check returns OK without credentials; all other endpoints correctly refuse access without the pilot password.
- All 8 tracked colleges present and queryable.
- Live-streamed answers arrive progressively (citations first, then text) through the proxy — the most fragile part of the stack, confirmed working on the real host.
- Full evaluation suite run against the deployed server: 33 / 33.

7. REMAINING BEFORE WIDER PILOT USE

Item	Note
Domain name + HTTPS	Currently plain HTTP on an IP address, so the password travels unencrypted — suitable for smoke-testing only. Adding a domain is a one-line change; certificates are then issued automatically.
Private link workstation → server	So new meetings ingested on the workstation publish straight to the pilot server.
Off-site backup copy	Deferred while the workstation remains a complete second copy of everything; ready to enable at no cost when needed.